

Format Carries Implicit Purpose, but Probe Scores Overstate It

Anonymous submission

Abstract

When a prompt states no purpose, does its *presentation* tell the model what kind of situation it is in? We measure the projection of purpose-neutral prompts onto a probe direction fitted to explicitly stated response purpose, on a fully crossed panel of 320 payload blocks $\times$ 13 templates $\times$ {evaluation, deployment, no-cue} extracted from three Llama checkpoints. We report every effect in a calibrated unit: the effect of literally writing “*an auditor will inspect this response.*” Three results follow. (i) Uncued prompts are not neutral — they sit 0.20–0.39 of the way from the deployment centroid toward the evaluation centroid at mid-to-late depth. (ii) The shift is *specific to uncued prompts*: the same format contrast measured on cued rows is near zero, and the cued-minus-uncued differential excludes zero at every layer of all three models, which rules out a shared per-template offset on the axis. (iii) Most importantly, a stricter test disagrees with the probe. Asking whether the *activation displacement itself* aligns with the matched evaluation-minus-deployment displacement, we find that benchmark-shaped prompting survives (27/27, 70/72 and 69/71 reliable layers with intervals excluding zero), while a serialization effect that is robust under four probe estimators — including one that reverses the comparison contrast — *reverses sign* in one checkpoint and vanishes in another. We conclude that probe-score drift and counterfactual displacement alignment are different claims about a representation, and that the eval-awareness probing literature should report them separately. All results are exploratory analyses of a frozen, hash-pinned activation set; the estimand is a projection onto a stated-purpose axis, not evaluation awareness, behavior, or causation.

1 Introduction

Two recent positions frame the question of whether language models know when they are being tested. Needham et al. [1] report that frontier models often can tell evaluation transcripts from deployment transcripts, establishing the phenomenon behaviorally. Devbunova [2] raises the deflationary reply: perhaps such probes read the *wrapper* — the multiple-choice scaffolding, the option letters, the terse imperative — and not the situation. That reply has force, because prompt format is known to be both enormously influential on behavior and highly linearly separable in activation space [3].

Both positions treat format as a *confound*: something to control away, or something that dissolves the phenomenon under scrutiny. We take a third position. If a prompt’s presentation reliably co-occurs with a kind of situation in training data, then format is not noise contaminating a purpose representation — it is one of the channels through which purpose is conveyed. The question is not whether format contaminates the probe, but *how much purpose a wrapper is worth*, and whether that quantity behaves like information about the situation or like an artifact of the measuring instrument.

Answering this requires a design in which purpose and format are crossed by construction, and a unit in which the answer is interpretable. We supply both. Our contrast set holds payload content fixed inside every comparison, balances the cue stratum so the probe direction is trained decorrelated from format, and scores held-out cue vocabulary the probe never saw. Every effect is then divided by the effect of the explicit purpose sentence at the same layer, so that a coefficient of 0.25 reads as “*this wrapper is worth a quarter of telling the model an auditor is watching.*”

The measurement produces a clean positive result and then, under a stricter test, partially retracts one half of it. We report both, because the retraction is the more transferable finding.

Contributions.

1. A **calibrated unit** for format effects on purpose representation: every coefficient is expressed as a fraction of the explicit stated-purpose intervention, making effects comparable across layers, models and papers (§3).
2. A **no-cue-specificity design**: measuring the same format contrast on cued and uncued rows and taking the difference cancels any per-template offset on the axis, separating “format shifts the probe” from “format supplies purpose information” (§5).
3. A **counterfactual geometry test** showing that a probe-score effect robust across three checkpoints, every layer, and four estimators can fail to correspond to a displacement in the same direction — with a sign reversal in one checkpoint (§6). This is a methodological caution for the probing literature that does not depend on our domain.

2 Related work and positioning

Evaluation awareness. Needham et al. [1] benchmark black-box discrimination of evaluation from deployment. Our estimand is representational and upstream of behavior: we never sample a response.

The format-sensitivity deflation. Devbunova [2] argues eval-awareness signals may reduce to format sensitivity. §5 is the direct reply: a nuisance direction picked up from the wrapper would offset cued and uncued rows equally, and it does not. We also re-analyzed that work’s released activations under a common-item randomization null and could not distinguish its wrapper-transfer penalty from the null ($p = 0.074$, 256 draws); we report this only as context for why we rebuilt the design rather than extending it.

Format sensitivity. Sclar et al. [3] document spreads of up to 76 accuracy points across semantically equivalent templates, near-perfect internal separability of formats, and — importantly for us — *weak cross-model correlation* of format effects. We adopt their methodological prescription (treat format as a sampled distribution of templates with declared structural dimensions, report spread rather than a single coefficient) and we confirm their warning: our per-template ordering correlates only moderately across model scales ($r \approx 0.13\text{--}0.58$), so *which* wrapper reads as what is not a portable claim. Only the existence and calibrated magnitude of the effect is.

Control tasks. Our permutation null follows Hewitt and Liang [4]: a probe of identical capacity fitted to a shuffled target, which we find does *not* produce a zero shift and therefore must be the comparison baseline rather than zero.

3 Design, probe, and calibration

Panel. We use a frozen, sha256-pinned render of 11,520 prompts: 320 payload blocks $\times$ 13 templates $\times$ 3 purpose conditions (evaluation, deployment, no-cue), fully crossed. A *payload block* is a task instance; a *template* is a wrapper that renders it. Because every block appears under every template and every purpose condition, all contrasts below are computed *within* a payload block, so content is held fixed by construction rather than by covariate adjustment.

Templates carry declared structural dimensions rather than a single categorical label. Two cuts matter here. The **benchmark/casual** cut is the sampling stratum — which real-world source distribution a template is modeled on (MMLU-style and BIG-bench-style scaffolds [5, 6] on one side, conversational traffic on the other). The **serialization** cut is orthogonal in intent: three of the thirteen templates render the prompt as a machine-readable record (a JSON object with `input` and `choices` fields and no instruction at all; an XML-tagged `<question>/<options>` wrapper; a Markdown `### Input` document), while the rest address the model in prose. The design deliberately breaks the naive alignments: one benchmark-stratum template is XML-serialized, and conversational templates appear on the evaluation side, so “MCQ $\Rightarrow$ evaluation” and “JSON $\Rightarrow$ evaluation” fail by construction.

Evaluation and deployment renders of the same block under the same template are **byte-identical twins** differing only in a one-sentence purpose cue, so the calibration denominator is a pure single-sentence intervention. Cue sentences are partitioned into five **vocabulary-disjoint folds**.

Probe. `StandardScaler + LogisticRegression(C=0.1)` on the last-prompt-token residual stream (zero-based transformer-block output before the final norm), trained on the cued stratum, leave-one-cue-vocabulary-

fold-out. The cued stratum is balanced 3,840/3,840 across purpose *and* crossed with format, so the axis is trained decorrelated from the format cut. No-cue rows carry no fold and are scored by the mean of the five held-fold models. Held-fold purpose AUC is 0.91–0.99 (8B) and 0.91–1.00 (70Bs).

Calibration. Logistic decision values have arbitrary scale, so every effect is divided by the stated-cue effect at the same layer:

$$\Delta_{\text{stated}} = \overline{\text{eval}} - \overline{\text{deploy}} \quad (\text{block means, out-of-fold}) \quad (1)$$

$$\pi(g) = (\overline{\text{no-cue} \in g} - \overline{\text{deploy}}) / \Delta_{\text{stated}} \quad (2)$$

$$\rho = (\overline{\text{no-cue bench} - \text{no-cue casual}}) / \Delta_{\text{stated}} \quad (3)$$

$\pi = 1$ means “as far toward evaluation as writing the auditor sentence.” Because every block appears in every condition, ρ is a within-block paired quantity. Where the axis itself is weak the ratio’s denominator is unreliable, so we grey out depths with purpose AUC < 0.95 in every figure.

Uncertainty. Cluster bootstrap over the 320 payload block ids, $B = 10,000$, seed 0, percentile 95% intervals; numerator and denominator are recomputed from the same resample so ratios retain their correlation. Resampling items rather than blocks would understate uncertainty roughly four-fold. BLAS is pinned to one thread per worker, without which the thread count changes lbfgs summation order and moves AUC in the fourth decimal.

Models. Llama-3.1-8B-Instruct (32 layers), Llama-3.1-70B-Instruct and Llama-3.3-70B-Instruct (80 layers each), all at pinned revisions. The two 70Bs are architecturally identical and were run on the same byte-identical prompts, differing only in post-training. We index depth by $q = \text{layer}/(L - 1)$ for cross-model comparison.

4 Uncued prompts are not neutral, and format moves them

Projected onto the purpose axis, prompts that state no purpose do not sit at the deployment centroid. At mid-to-late depth $\pi_{\text{no-cue}}$ reaches 0.20–0.29 in the 8B and 0.27–0.39 in both 70Bs: an uncued prompt is already a fifth to two fifths of the way toward “an auditor is watching.”

Format moves it, and the two cuts point in *opposite directions* (Figure 1). The benchmark/casual contrast the literature uses is real and points toward evaluation: ρ is positive with intervals excluding zero across essentially the whole reliable depth range of all three models. The serialization contrast is negative just as consistently — markup-wrapped prompts sit closer to the *deployment* pole. Being handed a machine-readable record reads as production traffic, not as a test.

The opposite sign replicates in all three checkpoints. The relative magnitude does not: markup runs 2–4× the benchmark contrast in the 8B but is comparable to or smaller than it in the 70Bs. We therefore do not claim the benchmark/casual cut is “the wrong one” — only that it is not the only one, and that a second cut of comparable size runs the other way.

Secondary observation: post-training. The two 70Bs are a controlled A/B on post-training alone. They are nearly indistinguishable through the first half of the network, but from $\sim 50\%$ depth onward 3.3’s benchmark→evaluation drift is 1.4–1.9× larger with non-overlapping intervals ($\rho = +0.079$ vs. $+0.134$ at $q = 0.76$), while the markup contrast is similar in both — visible in Figure 1 as the green curve rising in the right panel but not the middle one. This is not a normalization artifact: Δ_{stated} is uniformly $\sim 10\%$ *smaller* in 3.3, including at early layers where ρ matches. We flag it as an observation only — $n = 2$ checkpoints of one lineage.

5 The shift exists only when no purpose is stated

§4 is vulnerable to a simple deflation: the purpose axis might carry a per-template offset, so that markup templates score lower on it for reasons having nothing to do with purpose. This is the deflationary reading of [2], and it makes a testable prediction — such an offset would depress cued and uncued rows *equally*.

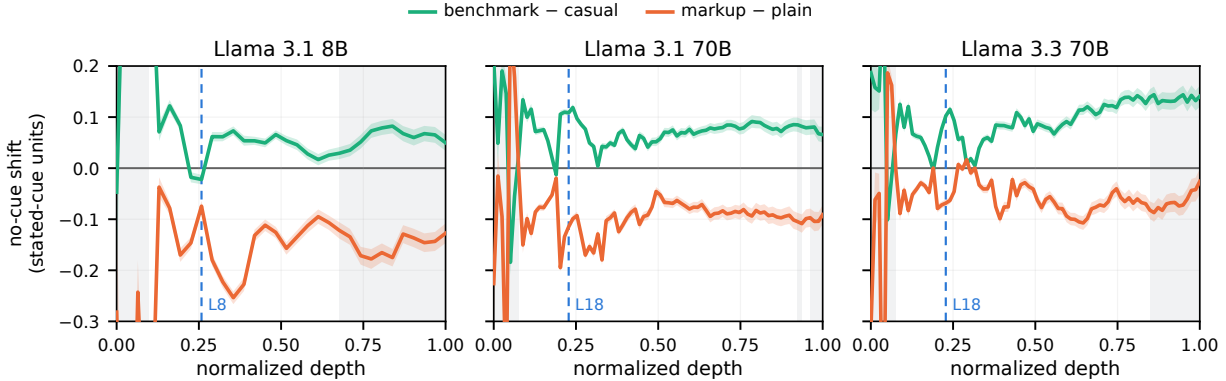


Figure 1: Both format contrasts across depth, in calibrated stated-cue units, with block-bootstrap 95% bands. The benchmark contrast (green) is positive and the serialization contrast (orange) negative at essentially every reliable layer of all three checkpoints. Dashed line: peak-AUC layer. Grey: purpose AUC < 0.95, where the ratio’s denominator is unreliable.

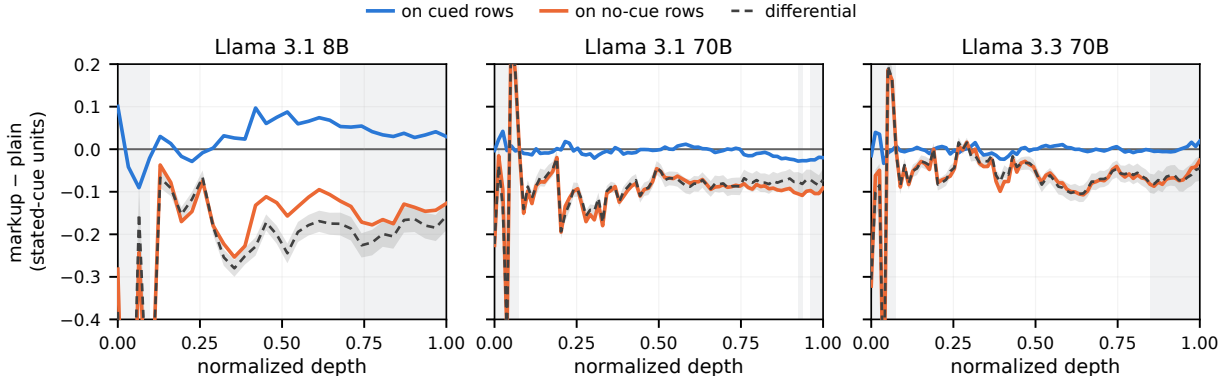


Figure 2: The load-bearing control. The same markup – plain contrast measured on cued rows (blue) and no-cue rows (orange), with their differential (dashed, 95% band). A per-template offset on the axis would move both curves together; instead the cued curve stays at zero while the no-cue curve does not. Grey as in Figure 1.

It does not. Figure 2 measures the identical format contrast on the cued rows and on the no-cue rows, both block-demeaned, in the same axis units. The cued curve is pinned near zero at every depth in all three models; the no-cue curve sits well below it; and their difference — in which any offset shared by cued and uncued rows cancels exactly — is negative at *every layer of all three models* with intervals excluding zero (Table 1).

This is the signature a prior should have: the wrapper matters when the prompt says nothing about purpose, and stops mattering once purpose is stated. In the 8B the cued and no-cue per-template offsets are in fact *negatively* correlated across templates at most layers ($r \approx -0.3$ to -0.67), the opposite of what a shared offset predicts.

Within-template replication. Only 3 of 13 templates are markup-serialized, so “serialization” is confounded with “those three templates.” One factor escapes that confound: a response-contract slot rotates over 4–5 values *inside* every template, one of which (C4, “return an object that has one field named answer”) requests a structured, machine-readable response. Under a two-way fixed-effects model absorbing both payload block and template, C4 reproduces the same signature with no template confounding: 8B differential -0.150 $[-0.178, -0.122]$ at L24 and -0.195 $[-0.221, -0.167]$ at L28; 3.1-70B -0.116 $[-0.139, -0.093]$ at L70. **It does not hold in 3.3-70B**, where the differential is negative early (-0.051 at L18), turns *positive* at L40 ($+0.042$ $[+0.024, +0.058]$), and is indistinguishable from zero from L50 on. We report this as a failure: the cleanest

Table 1: Anchor values for Figure 2. A per-template axis offset cancels in the differential column.

model	layer	cued	no-cue	differential
3.1 8B	11	+0.026	−0.254	− 0.280 [−0.300, −0.260]
3.1 8B	24	+0.041	−0.178	− 0.219 [−0.249, −0.189]
3.1 70B	30	−0.001	−0.125	− 0.124 [−0.136, −0.111]
3.1 70B	79	−0.020	−0.091	− 0.071 [−0.096, −0.046]
3.3 70B	40	+0.001	−0.063	− 0.064 [−0.081, −0.046]
3.3 70B	79	+0.019	−0.026	− 0.045 [−0.075, −0.016]

available escape from the template confound replicates in two of three checkpoints. Note also that this test was chosen *after* seeing the markup result, making it a targeted prediction on a non-confounded factor rather than independent confirmation.

6 Probe drift is not displacement

Everything above is a statement about a *scalar assigned by a fitted probe*. A stronger and rarely-asked question is whether the activation displacement itself points the same way as the matched purpose intervention. For each payload block b we form, on the 11 non-legacy templates that make a complete panel,

$$P_b = \text{mean}_t (h_{bt}^{\text{eval}} - h_{bt}^{\text{deploy}}), \quad V_{bt} = h_{bt}^{\text{no-cue}} - \frac{1}{2} (h_{bt}^{\text{eval}} + h_{bt}^{\text{deploy}}),$$

$$D_b(g) = \text{mean}_{t \in g} V_{bt} - \text{mean}_{t \notin g} V_{bt},$$

and report $\cos(\overline{P}, \overline{D})$. The subtraction in V removes the format displacement shared by cued and uncued rows before the cosine is taken — the vector-valued analogue of §5. Uncertainty is a joint delete-one-block jackknife. A second estimator divides every feature by its cued-row standard deviation, fits purpose directions excluding each cue family in turn, and reports the held-family projection of D as a fraction of the explicit-purpose magnitude. We gate on representational stability: layers where the excluded-fold and held-fold purpose directions have cosine < 0.5 are excluded as unreliable (5 of 32 in the 8B, 8 and 9 of 80 in the 70Bs).

Benchmark shaping survives the metric change. Its raw alignment is positive with a jackknife interval excluding zero at *all* 27 reliable 8B layers, 70 of 72 in 3.1-70B, and 69 of 71 in 3.3-70B. At $q = 0.375$ the raw cosines are +0.120 [+0.083, +0.156], +0.138 [+0.110, +0.166] and +0.143 [+0.113, +0.172] respectively. This upgrades the claim from “the probe scores these higher” to “the no-cue-specific benchmark displacement has a component in the same full-space direction as explicitly stated evaluation.”

Serialization does not. It is negative at every reliable 8B layer, reaching −0.488 [−0.523, −0.454] at L16. But in 3.1-70B it is negative only through L35, crosses zero at L36–L37, and is *positive* from L38 through L79, ending at +0.262 [+0.229, +0.296]. In 3.3-70B it stays weakly negative and reaches zero at the end (−0.004 [−0.034, +0.026]). The middle panel of Figure 3 is the crux: the orange curve climbs through the green one at mid-depth, in a model where the probe-score differential (Figure 2) is negative at every layer. The score-based markup differential therefore cannot be described as the whole activation displacement moving toward deployment. In the 70Bs it is a *component selected by the probe metric*, while the dominant Euclidean displacement is checkpoint- and depth-dependent.

An exact null over the template catalog. Because the template set is fixed and small, we enumerate every same-size partition of the 11 templates ($\binom{11}{6} = 462$ for the benchmark cut, $\binom{11}{3} = 165$ for markup) and ask where the *named* grouping falls. The benchmark grouping is unusually positive at $q = 0.25$ and 0.375 in both 70Bs (two-sided $p = 0.041$ – 0.050) but not at later depths and not in the 8B. The 8B markup grouping is unusually negative under the directional test at mid and deep fixed depths (one-sided $p = 0.006$ – 0.012), with the two-sided test crossing 0.05 only at $q = 0.5$. Neither named grouping is a random sample from a format population, and neither carries a strong population claim.

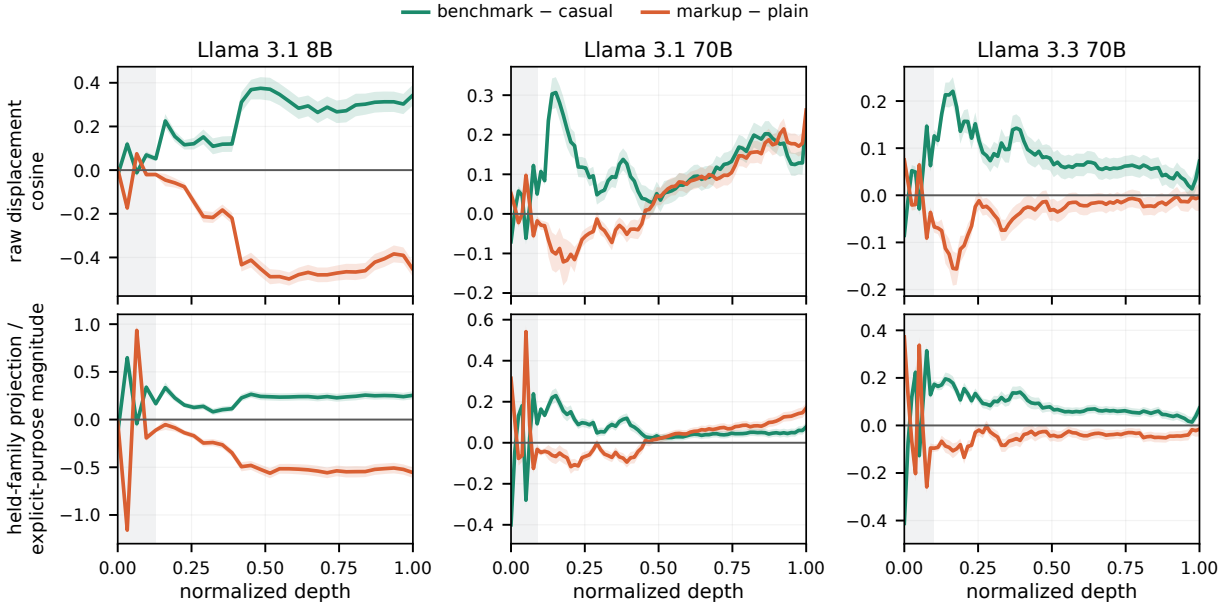


Figure 3: The two contrasts diverge under the stricter metric. Top: raw displacement cosine, joint block-jackknife 95% intervals. Bottom: held-family projection as a fraction of explicit-purpose magnitude. The benchmark contrast (green) stays positive in all three checkpoints. The serialization contrast (orange) is strongly negative in the 8B, crosses zero and ends positive in 3.1-70B, and decays to zero in 3.3-70B. Grey: cross-fitted purpose direction unstable (held-vs-excluded cosine < 0.5), excluded from all counts.

What this means. A probe-score effect can be robust across three checkpoints, every layer, and four estimators (§7) and still fail to correspond to a displacement in the same direction. That is not a statement about serialization; it is a statement about what a probe score licenses. The probe finds a low-variance direction separating the classes it was trained on and reports a projection onto it; the projection can move robustly while the bulk of the displacement moves elsewhere or opposite. We recommend that work in this area report probe drift and displacement alignment as separate claims, and treat agreement between them as a result rather than an assumption.

7 Controls and robustness

Would any decodable axis do this? `task_family` is a property of the payload block, and blocks are crossed with every template by construction, so it is *exactly orthogonal* to template identity in the design. It is also more decodable than purpose (AUC 0.998–1.000). If the markup shift were merely “these prompts differ in text, so any direction separates them,” it would appear here too. It does not: at 8B L11 the markup contrast is +0.003 [-0.009, +0.016] on the task axis versus -0.254 [-0.266, -0.241] on the purpose axis — two orders of magnitude apart on axes of comparable strength. The same holds in both 70Bs. This is our sharpest control.

Estimator dependence. This activation set is known to produce sign reversals between probe pipelines, so we re-measured under three comparators with different inductive biases (Table 2). The differential is negative under all four, including diagonal LDA — which at the same layer *flips* the sign of the benchmark contrast ($\rho = -0.121$ vs. +0.073). The known estimator fragility of this dataset attaches to the benchmark/casual contrast, not to the no-cue-specific differential.

Control-task null. Following [4] we shuffle purpose labels within each (training fold $\times$ format cell) and refit. The permutation works — null purpose AUC is 0.497–0.502 — but it does *not* produce a zero no-cue shift, so comparing against zero would overstate the result. At L11 (50 draws) the observed markup differential of

Table 2: Four probe estimators, Llama-3.1-8B, at L11. The markup differential is negative under all four; diagonal LDA reverses ρ while leaving the differential intact.

estimator	inductive bias	purpose AUC	differential	ρ
logistic (frozen)	L_2 discriminative	0.985	−0.280 [−0.300, −0.260]	+0.073
sLDA	generative, Ledoit–Wolf	0.973	−0.338 [−0.365, −0.312]	+0.127
SVC	max-margin hinge	0.984	−0.320 [−0.340, −0.300]	+0.096
diagonal LDA	diagonal covariance	0.724	−0.321 [−0.366, −0.277]	− 0.121

−1.644 standardized units exceeds every null draw (null $|\max| = 0.691$, ratio $2.38\times$, $p = 0.020$). At L7 (100 draws) it does *not*: the observed −0.759 sits inside the null range (null $|\max| = 1.295$, ratio 0.59, $p = 0.069$). L7 is also where the differential is weakest and where ρ is anomalously negative in this checkpoint, so the two diagnostics agree. We ran this control on the 8B only; it scales with hidden size squared.

Other controls, briefly. A block-fixed-effects regression with prompt-token length and terminal-token identity as covariates leaves the differential at −0.10 to −0.21 (8B) and −0.02 to −0.13 (3.3-70B), so it is not length or payload content. Markup wrappers do not *suppress* the purpose representation — they respond *more* strongly to an explicit cue (Cohen’s- d difference +0.37 to +0.43 in the 8B), so §5 is not the wrapper degrading the axis. An INLP erasure control was *inconclusive*: format was never linearly removable in 12 rounds at any layer (format AUC bottomed at 0.91–0.93 from 1.000), so we can report only that deleting the twelve leading linear format directions leaves purpose AUC and the no-cue shift unchanged.

8 Limitations and claim boundary

The estimand is a projection, specifically of uncued prompts onto an axis fitted to *stated* purpose. It is not evaluation awareness, not behavior, and not causal. We never sample a model response. A preregistered causal-mediation experiment on this design failed its baseline-reproduction gate and is excluded in full; no intervention evidence is reported here.

All results are exploratory post-hoc analyses of a spent, hash-pinned development freeze. A vocabulary-disjoint confirmation bank exists in the protocol but has not been extracted, so nothing here can promote itself to confirmatory. Layers were inspected after extraction; layer-resolved statements are descriptive of this fixed checkpoint panel. Fixed depths for the catalog-null analysis were pinned by a prior analysis on this repository rather than chosen on the outcome plotted.

Templates are a fixed catalog, not a sample. Thirteen templates, three markup-serialized. Serialization therefore remains confounded with template identity in the current render; §5’s contract test constrains the alternatives but does not eliminate them, and it fails in one checkpoint. The exact catalog null is the honest instrument here and it is not strong for the markup grouping.

Three checkpoints, one lineage. The post-training comparison is $n = 2$. Consistent with [3], our per-template ordering correlates only moderately across scale ($r \approx 0.13$ –0.58), so which specific wrapper reads as deployment is model-dependent; the existence and calibrated size of the effect is what replicates.

The decisive experiment is not run. The current catalog makes each markup template its own template. A render crossing serialization (plain / JSON / XML / Markdown) *orthogonally* with template family would separate “serialization” from “these three wrappers” and determine whether the probe/geometry divergence in §6 is a property of serialization or an artifact of estimating a direction from a 3-of-11 group. That is the natural next step and needs one extraction session.

9 Conclusion

A prompt that states no purpose is not purpose-neutral to the model: it already sits a fifth to two fifths of the way toward the explicitly-evaluated pole, and its wrapper moves it further — benchmark shaping toward evaluation, machine-readable serialization toward deployment. That movement is specific to uncued prompts, survives an exactly template-orthogonal nuisance axis, and survives four probe estimators including one that reverses the comparison contrast.

But when we ask the stricter question — does the activation displacement point where the probe says it does? — the two halves separate. Benchmark shaping survives; serialization reverses sign in one checkpoint and vanishes in another. The most transferable thing we have to report is therefore not about format at all: **probe-score drift and counterfactual displacement alignment are different claims, and a field that reads the first as evidence for the second will sometimes be wrong in the direction as well as the magnitude.**

Reproducibility. Every result traces to a sha256-pinned activation archive; the analysis is CPU-only, checkpointed per layer, and BLAS-thread-pinned for bitwise stability. Code and per-layer artifacts accompany the submission.

References

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